

DAVID GARGAN

PRINCIPAL AI SYSTEMS ARCHITECT | ENTERPRISE KNOWLEDGE & GOVERNANCE | OPENTEXT ECM

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PROFESSIONAL SUMMARY

Principal-level systems architect with 25+ years of experience designing and delivering enterprise information systems, with recent work centered on applied AI, LLM operations, knowledge extraction, evaluation, provenance, and multi-agent orchestration. Combines hands-on engineering with deep experience in information architecture, governance, security, records management, workflow, regulated environments, and large-scale migrations. Proven record across government, defense, aerospace, energy, pharmaceutical, and enterprise programs where correctness, traceability, and operational reliability are critical.

AI, EVALUATION & TECHNICAL FOCUS

LLM Evaluation & Review • Agentic / Multi-Agent Systems • Applied ML Operations • RAG & Semantic Retrieval • PostgreSQL / pgvector • Dataset & Knowledge Extraction • LoRA / QLoRA Workflows • AI Governance & Provenance • Human-in-the-Loop Quality Controls • Python • FastAPI • SQL • C# • REST APIs • Docker • Git • Linux • OpenText Content Server / Livelink • xECM • Records Management • WebReports • Workflow • Taxonomy / Information Architecture

PROFESSIONAL EXPERIENCE

Avanti AI Innovators / Avanti Complex | Founder & Principal AI Systems Architect

Mar 2024 - Present

- **Architected VALESKA** as an enterprise AI knowledge and institutional-memory system, using PostgreSQL as the authoritative metadata/provenance layer, pgvector for semantic retrieval and domain discovery, and a governed graph model for entities, relationships, and cross-document assertions.
- **Designed and implemented VALESKA Commons**, a live API-centered coordination layer for independent AI agents and coding harnesses, with task/decision records, checkpoints, pre-action interlocks, conflict-aware work ownership, provenance, and an evolving event/mailbox model.
- **Architected DaveRouter**, a policy-driven agent execution gateway that converts natural-language objectives into governed work, compiles bounded context, selects among Claude, Codex, Grok, Hermes, local inference, and cloud APIs, and reconciles results against Commons and Git state.
- **Developed local LLM training and evaluation workflows** for a 24 GB GPU environment, including dataset preparation, LoRA/QLoRA planning, retrieval experiments, model comparison, review escalation, and token-efficient multi-model routing.
- **Designed applied computer-vision workflows** for a Python-based mobile PWA that captures estate-sale inventory, localizes/segments objects, routes image regions through vision models, and integrates human review, labeling, and pricing operations.
- **Led design of AI-assisted trust and identity systems** integrating certificate verification, X.509 workflows, blockchain-backed audit concepts, secure data exchange, and privacy-preserving verification techniques.

Stellar Services Inc. | Team Lead / Solution Architect - OpenText Content Server 20.x

Dec 2021 - Mar 2024

- Led architecture for Texas Department of Public Safety replacement of legacy Document Manager functionality, including CIS modeling, Records Management, Categories & Attributes, C#/web-service integrations, workflows, system sizing, and migration coordination.
- Served as principal liaison to Regulatory Services; authored business requirements and functional/technical design documentation and developed advanced WebReports to monitor workflows processing approximately 800 items per day.
- Acted as an AI advocate, identifying opportunities to apply emerging AI capabilities to enterprise content, classification, retrieval, and workflow operations.

TransCanada Energy | OpenText Technical Consultant

Jul 2021 - Sep 2021

- Applied complex enterprise security models and resolved OpenText Object Importer issues during a FileNet-to-Content Server 16.2.11 migration involving millions of tightly coordinated documents.

Sempra Energy | OpenText Technical Consultant

Feb 2021 - Apr 2021

- Supported modernization of an OpenText CS16 environment with emphasis on taxonomy, Records Management, WebReports, and enterprise information structure.

OpenText | Technical Consultant

Jul 2017 - Apr 2020

- Selected by OpenText for direct client work after serving as a partner/independent consultant; certified through OpenText programs and delivered implementations across government, pharma, energy, and transportation.
- Designed xECM Records Management solutions on Microsoft 365 and Content Server 16.2; modeled content, information, security, retention, and physical records requirements for regulated environments.
- Automated repetitive XML-control operations with C# and VBA and taught OpenText employees and customers in classroom, virtual, and on-premises settings.

AI STORYTELLING & VIDEO PORTFOLIO

AI storyteller and video creator behind Transmissions from the Timeline, including Cat & Daniel: Collapse Protocol, serialized science-fiction stories, videos and Shorts.

Portfolio: youtube.com/@TransmissionsfromtheTimeline

Storytelling & Media Lab: storylab.avanticomplex.com

EARLIER ENTERPRISE ARCHITECTURE & CONSULTING EXPERIENCE

Avanti Complex / IQ Business Group | Independent Contractor / Solution Architect **Apr 2016 - Jul 2017**

- Delivered C# console, MVC, web-service, workflow, migration, and Content Server integration solutions for the Philadelphia District Attorney, Defense Logistics Agency, U.S. Army, and Kootenai and Salish Nation.
- Led strategy and migration work for highly customized platforms dependent on aging legacy systems; integrated Content Server with Cordys and custom front-end process controls.

AliTek Consulting | Senior Consultant **Jun 2013 - Feb 2015**

- Permanent member of a TVA migration team; applied process and programming improvements that cut legacy transport time by 80% while maintaining near-100% accuracy.

Limited Brands | Technical Consultant / Livelink System Administrator **Oct 2009 - Jun 2013**

- Administered a high-visibility enterprise Livelink environment and managed/budgeted an upgrade from Livelink 9.7.5 to Content Server 10.
- Created a Knowledge Manager operating model used across 10 time zones to distribute administrative workload and implemented Records Management within the Tax Department.

Stelix | Senior Consultant **Apr 2008 - May 2009**

- Designed and integrated Records Management and Advanced Workflow supporting a demanding 21 CFR Part 11-compliant pharmaceutical solution for a new product launch.

BearingPoint | Senior Consultant **Jan 2005 - Mar 2008**

- Created VALESKA (Value Stream Knowledge Architecture), a Lean-based approach to CIS and knowledge architecture; received a grant to develop the methodology into a product.
- Participated in development of an SAP Composite Application; team received a BearingPoint Innovation Award.

NASA Dryden Flight Research Center | Livelink Administrator / Developer **2001 - 2005**

- Developed an ASP-based, data-driven maintenance tracking system used across multiple research-center asset groups; success led to administration/development responsibility for two Livelink DMS environments on Windows and Solaris.

NASA Ames Research Center | Solaris / Windows Administrator & Programmer **1999 - 2001**

- Supported research networking and video-transmission experiments; maintained network infrastructure and experimental systems, including field deployments supporting White Sands and Biggs Army Airfield test programs.

U.S. Navy | Linguistics Supervisor / Combat Systems Specialist **1984 - 1994**

- Led technical and operational teams in assignments spanning missile-tracking radar, Aegis combat information systems, combat operations, and the transfer of four U.S. Navy frigates to the Brazilian Navy.
- Certified Portuguese translator/interpreter; completed service as a First Class Petty Officer.

EDUCATION & CERTIFICATIONS

- University of South Florida - B.A., International Studies
- OpenText Certified Content Server 16.2 Systems Administrator; OpenText Content Server 16 Certified Consultant
- OpenText Certified xECM for Microsoft Office 365 16.2 System Administrator
- Content Server Records Management; OpenText Content Server Builder Fundamentals
- CompTIA Security+; NASA Network and Internet Security; Microsoft Systems Engineer (MCSE); MCP+Internet
- Department of the Navy Language Certification - Portuguese (Brazilian and Portugal)

SELECTED ACHIEVEMENTS, LANGUAGES & PRIOR CLEARANCES

- BearingPoint Innovation Award (2007); Navy leadership training; Navy Good Conduct Medal; Navy Expeditionary Medal; Armed Forces Expeditionary Medal.
- Letters of Commendation/Appreciation for work supporting the transfer of U.S. Navy frigates to the Brazilian Navy.
- Languages: English (native); Portuguese (fluent, certified translator/interpreter); Spanish (conversational).
- Prior security clearances: DoD Secret; TVA Restricted. Clearance status not represented as current.